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CS-2614-Computer Organization

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1. Show the value of all bits of a 12-bit register that hold the number equivalent to decimal 215

a) Binary:

$$215_{10} \rightarrow 11010111_2$$

c) Binary-coded Hexadecimal

$$215_{10} \rightarrow D7_{16} \rightarrow 1101\ 0111_2$$

↓ 12 bit

$$D \rightarrow 1101_2 \quad 7 \rightarrow 0111_2$$

b) Binary-coded octal:

$$215_{10} \rightarrow 327_8 \rightarrow 011\ 010\ 111_2$$

↓ 12 bit

$$3 \rightarrow 011_2 \quad 2 \rightarrow 010_2 \quad 7 \rightarrow 111_2$$

d) Binary-coded decimal

$$215_{10} \rightarrow 0010\ 0001\ 0101_2$$

↓ 12 bit

$$2 \rightarrow 0010_2 \quad 1 \rightarrow 0001_2 \quad 5 \rightarrow 0101_2$$

3. Obtain the 10's complement of the following six digital decimal numbers:

a) 123400

Subtract each digit from 9 → 876099

$$\text{add } 1 \rightarrow 876100$$

c) 100000

Subtract → 899999

$$+1 \rightarrow 900000$$

b) 090657

Subtract → 909342

$$+1 \rightarrow 909343$$

d) 000000

Subtract → 999999

$$+1 \rightarrow 1000000$$

↳ 6 digits only → 000000

5. Convert numbers to gray code:

a) 01001

$$01001 \rightarrow \text{gray code} \rightarrow 01101$$

b) 43₁₀ → 10101₂

$$10101 \rightarrow \text{gray code} \rightarrow 111110$$

c) 3A₁₆ → 001111010₂

$$001111010 \rightarrow \text{gray code} \rightarrow 001000111$$

d) 11111₂

$$11111 \rightarrow \text{gray code} \rightarrow 10000$$

e) 55₈ → 101101₂

$$101101 \rightarrow \text{gray code} \rightarrow 111011$$

f) 101001₂

$$101001 \rightarrow \text{gray code} \rightarrow 111101$$

g) 110101₂

$$110101 \rightarrow \text{gray code} \rightarrow 101111$$

7. Find the binary number of 42 using 8-bit 2's complement representation

$$42_{10} \rightarrow 00101010_2 \quad 42 \text{ is positive so no change needed}$$

9. Consider this two's complement binary number: 10011110
What is its decimal value?

$$10011110 \xrightarrow{\text{invert}} 01100001$$

num is negative

$$01100001 + 1 \rightarrow 01100010 \rightarrow 48_{10} \rightarrow -48_{10}$$

12. Perform the following operations

a) 10110 + 11001

$$10110_2 = -10_{10} \quad 11001_2 = -7_{10}$$
$$-10 + (-7) = -17_{10}$$

b) 11001 + 11000

$$11001_2 = -7_{10} \quad 11000_2 = -8_{10}$$
$$-7 + (-8) = -15_{10}$$

14. Which operation generates overflow

4 bits is from -8 to 7 outside of that is overflow

a) 0011 + 1100

$$3 + (-4) = -1_{10}$$

no overflow

b) 0111 + 1111

$$7 + (-1) = 6_{10}$$

no overflow

c) 1110 + 1000

$$-2 + (-8) = -10_{10}$$

overflow

d) 0110 + 0010

$$6 + 2 = 8_{10}$$

overflow

17. IEEE representation for 35

normalize → 1.1₂ = 1.1₂ × 2⁵

Sign bit (0) = positive

Exponent: 127 + 5 = 132 = 10000100₂

Mantissa: 011001001100000000000000

Combine: 01000111011001001100000000000000

19. Show IEEE representation

a) 356.75

normalize: 10100100.11₂ → 1.0110010011₂

Sign: 0 → positive

Exponent: 127 + 8 = 135₁₀ → 10000111₂

Mantissa: 011001001100000000000000

Combine: 01000111011001001100000000000000

21) X+Y result in IEEE

$$X = 1.01101111 \times 2^8$$
$$Y = 0.100010010100010000000000$$

$$137 - 127 = 10 \rightarrow 1.00101001 \times 2^{10}$$

$$1.00101001 \times 2^{10} = 100101001 \times 2^6 = 593_{10}$$

$$X+Y: -592.97865625 + 593 = 0.02134375$$

$$0 \rightarrow \text{positive}$$

$$\text{normalize } 10010000.11111111 \times 2^4$$

$$\rightarrow 1.0010000111111111 \times 2^8$$

$$\rightarrow 100100001111111111 \times 2^4$$

$$\rightarrow 1001000011111111110000$$

22. add the binary numbers

$$1.000 \times 2^1 + 1.110 \times 2^{-1}$$

$$1.000 \times 2^1 + (-1.110 \times 2^{-1})$$

$$1.000 + (-0.1110) = 0.001$$

$$1.000 \times 2^{-4}$$

$$1.010 \times 2^{-2} + 1.110 \times 2^{-3}$$

$$1.110 \times 2^{-2}$$

$$1.010 + 1.110 = 10.001$$

$$\text{normalize} \rightarrow 1.0001 \times 2^{-1}$$

23. Add numbers. Convert to binary first

$$5.75_{10} + 2.25_{10}$$

$$\rightarrow 101.11_2 + 10.01_2 = 1000.00_2$$

$$\text{normalize: } 1.00000 \times 2^3 \rightarrow 8_{10}$$

$$3.5_{10} + 2.5$$

$$11.1_2 + 0.1_2 = 11.11_2$$

$$\text{normalize} \rightarrow 1.111 \times 2^3 \rightarrow 3.75_{10}$$

2. Write your name in ASCII using an 8-bit code with the leftmost bit always 0. Include a space between names & a period after a middle initial

$$C \rightarrow 01000011_2$$

$$O \rightarrow 01101111_2$$

$$I \rightarrow 01101100_2$$

$$b \rightarrow 01100010_2$$

$$y \rightarrow 01111001_2$$

$$O100011 \quad 01101111 \quad 01101100 \quad 01100010 \quad 01111001 \quad 01010000 \quad 00101110$$

$$01000110 \quad 01100100 \quad 01101001 \quad 01100111 \quad 01101111 \quad 01011110$$

$$4. \text{ Perform the subtraction with the following unsigned binary numbers by taking the 2's complement of the subtrahend}$$

$$a) 11010 - 10000$$

$$10000 \rightarrow 0111 + 1 \rightarrow 10000$$

$$11010 - 10000 \rightarrow 1010$$

$$\text{discard overflow } 01010_2$$

$$c) 100 - 110000$$

$$110000 \rightarrow 011111 + 1 \rightarrow 0110000$$

$$000100 + 010000 \rightarrow 010100$$

$$P \rightarrow 01010000_2$$

$$O \rightarrow 00101110_2$$

$$F \rightarrow 01000110_2$$

$$r \rightarrow 01110010_2$$

$$i \rightarrow 01101001_2$$

$$S \rightarrow 01110011_2$$

$$O \rightarrow 01101111_2$$

$$n \rightarrow 01101110_2$$

$$b) 11010 - 11011$$

$$01101 \rightarrow 10010 + 1 \rightarrow 10011$$

$$11010 + 10011 \rightarrow 101101$$

$$\text{Discard} \rightarrow 01101$$

$$d) 1010100 - 1010100$$

$$1010100 \rightarrow 01010111 + 1 \rightarrow 0101100$$

$$1010100 + 0101100 = 10000000$$

$$\text{discard overflow} \rightarrow 00000000$$

6. Convert the following numbers represented in gray code to the bases shown.

a) 1001 to binary

$$1001 \rightarrow 1110_2$$

b) 10011 to hex

$$10011 \rightarrow 1D_{16}$$

c) 10001 to octal

$$10001 \rightarrow 36_8$$

d) 11111 to decimal

$$11111 \rightarrow 21_{10}$$

8. Find the binary number of -42 using 8-bit 2's comp rep.

$$42_{10} \rightarrow 00101010_2$$

$$\text{invert} \rightarrow 11010101$$

$$\text{add } 1 \rightarrow 11010101 + 1 = 1101100_2$$

10. Find the 8 bit 2's complement binary number rep. of 137

$$137_{10} \text{ is outside the } -128 \text{ to } 127 \text{ range}$$

11. Represent the following using 8-bit 2's complement rep.

a) -47₁₀

$$47_{10} \rightarrow 00101111_2 \xrightarrow{\text{invert}} 11010000_2 \xrightarrow{+1} 11010001_2$$

b) -123₁₀ → -83₁₀

$$83_{10} \rightarrow 01010011_2$$

$$\text{invert} \rightarrow 10101100$$

$$+1 \rightarrow 10101101$$

c) 5C₁₆ → 92₁₀

$$92_{10} \rightarrow 01011100_2$$

16. Add numbers, indicate overflow if signed/unsigned

a) 12₁₀ + 8₁₀ 5 bit

$$12_{10} \rightarrow 01100_2 \quad 8_{10} \rightarrow 01000_2$$

$$01100_2 + 01000_2 = 10100_2 \rightarrow 12_{10}$$

$$\text{signed} \rightarrow \text{no overflow} \quad \text{unsigned} \rightarrow \text{overflow}$$

$$-12 \neq 20 \rightarrow \text{error}$$

b) A₁₆ + 24₁₆ 6 bit

$$A_{16} \rightarrow 10_{10} \rightarrow 00101_2$$

$$24_{16} \rightarrow 36_{10} \rightarrow 0010100_2$$

$$00101_2 + 0010100_2 = 10111_2$$

$$\text{unsigned} \rightarrow \text{no overflow}$$

$$\text{signed} \rightarrow \text{overflow} \rightarrow \text{same reason}$$

$$\text{Sign: } 1 \text{ positive}$$

$$\text{Exp: } 2 \dots 3 = 27_{10} - 127_{10} = -100$$

$$\text{Mantissa: } 2 \dots 3 \text{ normalize} \rightarrow 1.0110101$$

$$\text{Combine: } -1.0110101_2 \times 2^{-100} \rightarrow -7.89 \times 10^{-31}$$

$$d) 1.00000000 \times 11101100 \dots$$

$$\text{Sign: } 1 \text{ negative}$$

$$\text{Exp: } 0_{10} \rightarrow M \neq 0 \Rightarrow \text{Denormalized number}$$

$$\text{Mant: } -0.011011_2 \times 2^{16}$$

$$\text{Combine: } -1.12 \times 10^{16}$$

20. IEEE to decimal

$$a) 01111111000010001100000000000000$$

$$\text{Sign: } 0 \text{ positive}$$

$$\text{Exponent: } 155$$

$$\rightarrow \text{NoN}$$

$$b) 01011001010010000000000000000000$$

$$\text{Sign: } 0 \text{ positive}$$

$$\text{Exponent: } 10111001 = 185_{10} - 127 = 58$$

$$\text{Mantissa: } 23\text{-bits normalize} \rightarrow 1.01001001_2$$

$$\text{Combine: } 1.01001001_2 \times 2^{58} \rightarrow \approx 1.267 \times 10^{17}$$

24. Write parity bit for the following nouns

a) 1100

$$\text{even par: } 2 \text{ 1s so pbit} = 0$$

$$\text{odd par: pbit} = 1$$

$$\text{even parity} = 11000$$

$$\text{odd parity} = 11001$$

b) 1000

$$\text{even parity: add \# of 1s pbit} = 1$$

$$\text{odd parity: pbit} = 0$$

$$\text{even par: } 10001$$

$$\text{odd par: } 10000$$

c) 0000

$$\text{even num 1s(0) so:}$$

$$\text{even pbit} = 0 \rightarrow 00000$$

$$\text{odd pbit} = 1 \rightarrow 00001$$

d) 1101

$$\text{add \# of 1s(3)}$$

$$\text{even pbit} = 1 \quad 11011$$

$$\text{odd pbit} = 0 \quad 11010$$